

Analysis of Statistical Intervals

1. Confidence Interval

An interval estimate for a population parameter is called confidence interval. A confidence interval bounds population or distribution parameters.

2. Tolerance Interval

Tolerance interval bounds a selected proportion of distribution.

3. Prediction Interval

Prediction interval bounds future observations from the population of distribution.

Confidence Interval

I Confidence interval on the mean of a normal distribution, variance known.

Suppose $x_1, x_2, \dots, x_n$ be a random sample of normal population with unknown mean μ and known variance σ^2 . The sample mean $\bar{x}$ is normally distributed with mean μ and variance σ^2/n , then

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

A confidence interval estimate for μ is of the form $l \leq \mu \leq u$, where 'l' and 'u' are the 'n' points of the sample data.

ie, $P(l \leq \mu \leq u) = 1 - \alpha$ where $0 \leq \alpha \leq 1$

Here 'l' is the lower confidence limit and 'u' is the upper confidence limit

$(1 - \alpha)$ is the confidence coefficient.

Confidence interval on μ is given by

$$\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where $\bar{X}$ - sample mean

$Z_{\alpha/2}$ - critical value

$\frac{\sigma}{\sqrt{n}}$ - standard error

$Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$ - marginal error

Q1 - Given $n=50$, $\bar{x} = 69$, $\sigma = 2$. Calculate 95% confidence interval (CI) for the given data.

We have to find CI for μ .

Given $n=50$, $\bar{x} = 69$, $\sigma = 2$.

$$95\% \text{ CI} \Rightarrow 100(1-\alpha)\% = 95\%$$

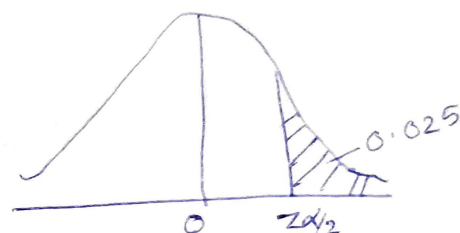
$$1-\alpha = \frac{95}{100}$$

$$\therefore \alpha = 1 - .95 = 0.05$$

$$\text{ie } \alpha = 0.05$$

$$\alpha/2 = \frac{0.05}{2} = \underline{\underline{0.025}}$$

$$P(Z > Z_{\alpha/2}) = 0.025$$



From tables,

$$P(Z < Z_{\alpha/2}) = 0.5 - 0.025 = 0.475$$

From tables,

$$Z_{\alpha/2} = 1.96$$

Confidence Interval is given by,

$$\bar{x} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$69 - 1.96 \times \frac{2}{\sqrt{50}} \leq \mu \leq 69 + 1.96 \times \frac{2}{\sqrt{50}}$$

$$\underline{\underline{68.44 \leq \mu \leq 69.55}}$$

Q. 2:- 10 measurement of impact energy of specimen are as follows: 64.1, 64.7, 64.5, 64.6, 64.5, 64.3, 64.6, 64.8, 64.2, 64.3. Assume that impact energy is normally distributed with $\sigma = 1$. Calculate 99% confidence interval on μ .

Given, $n = 10, \sigma = 1, \bar{x} = 64.46$.

$$99\% \text{ CI} \Rightarrow 100(1-\alpha)\% = 99\%$$

$$1-\alpha = \frac{99}{100}$$

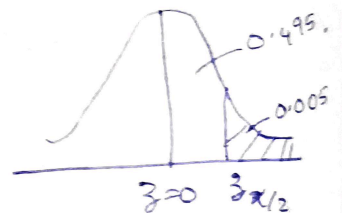
$$\alpha = \underline{\underline{0.01}}$$

$$\alpha/2 = \frac{0.01}{2} = \underline{\underline{0.005}}$$

$$P(Z > Z_{\alpha/2}) = 0.005$$

$$P(0 < Z < Z_{\alpha/2}) = 0.5 - 0.005 = 0.495$$

From tables, $Z_{\alpha/2} = 2.58$



$\therefore$ Confidence Interval is given by

$$\bar{x} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$64.46 - 2.58 \cdot \frac{1}{\sqrt{10}} \leq \mu \leq 64.46 + 2.58 \cdot \frac{1}{\sqrt{10}}$$

$$\underline{\underline{63.4864 \leq \mu \leq 65.27}}$$

Large sample confidence interval for μ

When 'n' is very large, the quantity $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ has an appropriate standard normal distribution, then the confidence interval is,

$$\bar{X} - Z_{\alpha/2} \cdot \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

is a large confidence for μ with confidence level of $100(1-\alpha)\%$.

II Confidence Interval on The mean of normal distribution with variance unknown.

t distribution

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution with unknown mean μ and unknown variance σ^2 , the random variable t ,

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $(n-1)$ degrees of freedom.

Q: Determine the value for 't' percentile.

a) Confidence level 95%, degree of freedom 12, two-tailed test.

$$100(1-\alpha)\% = 95\%$$

$$1-\alpha = \frac{95}{100}$$

$$\alpha = 0.05$$

$$\text{Given d.f} = 12$$

$$\therefore t = 2.179 \quad (\text{from t-distribution table})$$

- b. Confidence level = 99%, Degree of freedom = 13,
Two-tailed test.

$$CI, 100(1-\alpha) \% = 99\%$$

$$1-\alpha = .99$$

$$\alpha = 0.01$$

$$\alpha/2 = 0.005$$

$$\text{Deg. freedom} = 13$$

$$\therefore t = \underline{\underline{3.012}}$$

- c. Confidence level = 95%, Degree of freedom = 14,
One tailed test.

$$CI = 95\%$$

$$100(1-\alpha) \% = 95\%$$

$$1-\alpha = .95$$

$$\alpha = 1 - 0.95 = 0.05$$

$$\alpha/2 = 0.025$$

$$\text{Deg. of freedom} = 14$$

$$\therefore t = \underline{\underline{2.145}}$$